

Digitaler Materialpass

ID DMP-METAL-003 - Version 1.0.0
Ausstellungsdatum 2025-05-15 - Zertifikatstyp EN 10204 3.1

Hersteller		Kunde		Geschäftsvorgang							
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Precision Aerospace Inc. Aviation Boulevard 789 Mountain View 94043 US materials@precision-aero.example.com		Bestellung PO-34567 · 500 kg Pos. 3 · 2025-04-25 Lieferung DN-89012 · 500 kg Pos. 1 · 2025-05-14							
Produkt											
Produktname	Aluminum Alloy 7075-T6		Name (AA)	7075-T6							
Chargen-ID	H-43210-01		Nummer (UNS)	A97075							
Oberflächenzustand	Rolled		Form	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm							
Gewicht	500 kg		Produktnorm	AMS 4045 (2023)							
Produktionsdatum	2025-05-12										
Ursprungsland	DE										
Chemische Analyse											
Schmelznummer H-43210 - Schmelzverfahren VAR - Gießverfahren VacuumCasting - Gießdatum 2025-05-10 - Probenort Ladle											
Symbol	Al	Zn	Mg	Cu	Cr	F1					
Einheit	%	%	%	%	%						
Min		5.1	2.1	1.2	0.18	0.18					
Max		6.1	2.9	2.0	0.28	0.28					
Ist-Wert	89.7	5.6	2.4	1.5	0.22	0.22					
F1 = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.22											
Mechanische Eigenschaften											
Tensile Strength ¹	Symbol	1	2	3	Durchschnitt	Min	Max	Std.-Abw.	Spez. Min	Methode	Status
Wert [MPa]	Rm	570	572	574	572	570	574	2	530	ASTM E8	✓
0.2% Yield Strength ¹	Symbol	1	2	3	Durchschnitt	Min	Max	Std.-Abw.	Spez. Min	Methode	Status
Wert [MPa]	Rp0.2	503	505	507	505	503	507	2	480	ASTM E8	✓
Elongation ¹	Symbol	1	2	3	Durchschnitt	Min	Max	Spez. Min	Methode	Status	
Wert [%]	A	10.8	11.2	11	11	10.8	11.2	10	ASTM E8	✓	
¹ 3 specimens tested											
Physikalische Eigenschaften											
Eigenschaft	Symbol	Ist-Wert	Soll/Min	Maximum	Methode	Status					
Density	ρ	2.81 g/cm³	2.81 g/cm³		ASTM B311	✓					
Coefficient of Thermal Expansion Temperature range: 20-100°C	α	23.4 10 ⁻⁶ /K	23.5 10 ⁻⁶ /K		ASTM E228	✓					
Thermal Conductivity At 25°C	λ	130 W/(m·K)	120 W/(m·K)		ASTM E1461	✓					
Specific Heat Capacity At 25°C	cp	862 J/(kg·K)	860 J/(kg·K)		ASTM E1269	✓					
Electrical Resistivity At 20°C	ρe	0.0538 μΩ·m		0.0550 μΩ·m	ASTM B193	✓					
Poisson's Ratio	ν	0.33	0.33		ASTM E132	✓					
Melting Range	Tm	477 - 635 °C	475 - 635 °C		ASTM E1142	✓					
Relative Magnetic Permeability	μr	1.00002		1.00010	ASTM A342	✓					
Surface Roughness	Ra	0.8 μm		1.6 μm	ISO 4287	✓					
Emissivity At 25°C	ε	0.09		0.11	ASTM E408	✓					
Surface Tension	γ	0.875 N/m	0.870 N/m		ASTM D971	✓					
Diffusion Coefficient	D	0.0000000023 m²/s	0.0000000022 m²/s		ASTM E1559	✓					
Purity						ISO 4967 method B					
Plate II, inspected area approx. 200 mm²											

Inclusion type	A fine	A thick	B fine	B thick	C fine	C thick	D fine	D thick	DS
Max	2.5	2	2	1	1.5	1	1.5	1	1.5
Wert	1	1	1.5	0.5	0.5	0	1	0.5	0.5

Validierung

We hereby certify that the material described above has been manufactured and tested in accordance with AMS 4045 and the specified test methods. All results are within the specified limits.

Validiert von **Elsa Müller**, Materials Engineer, Quality Assurance - 2025-05-15